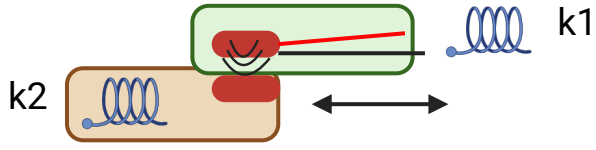
"2-aixs Wireless" E-skin Bend Sensors -
Taking Advantage of Resonant Frequency Shifts from Mutual Inductance + Difference in Spring Stiffness's

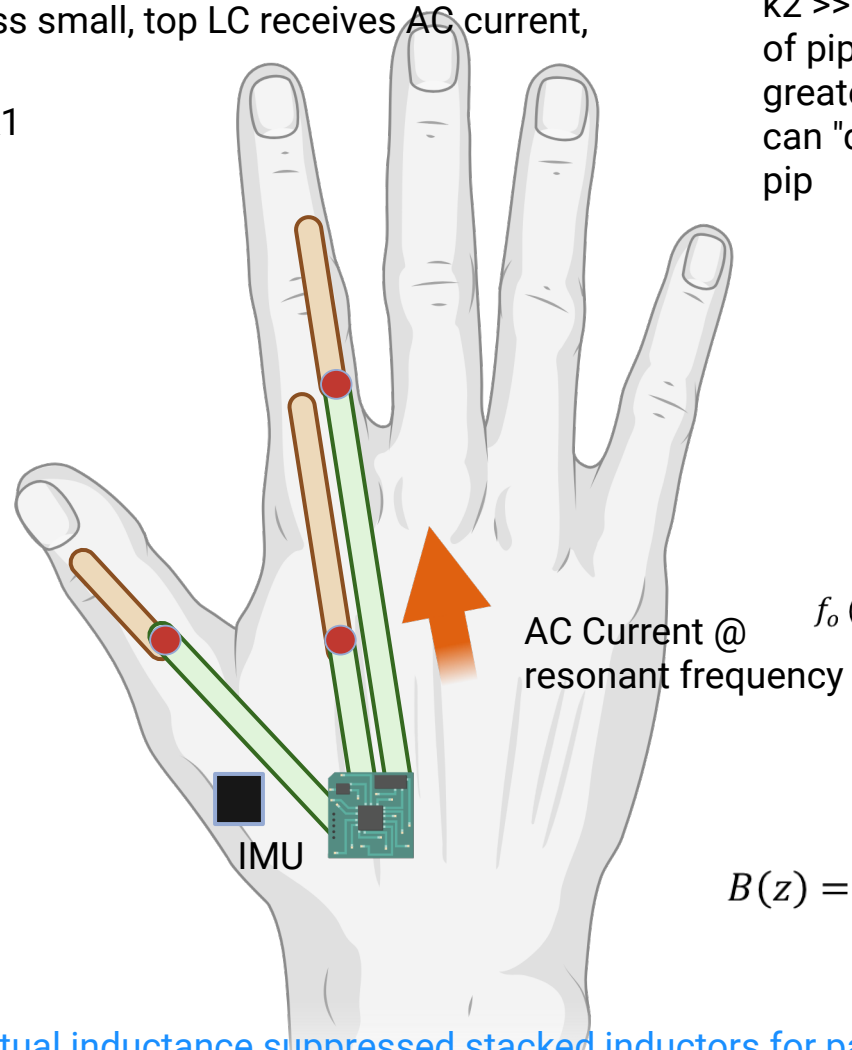
thin kapton tape with metal plated traces or threaded wires to tape to controller? Stiffness small, top LC receives AC current, creates Mag field



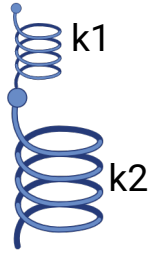
thin rubber sticker with coil/cap embedded. This part stuck to skin. As knuckle bends, it pulls bottom coil away from top coil, changing resonant frequency magnitude. Bottom LC is being mutually inducted

2x (as small as 3mm in dia) Inductor in parallel with capacitor LC resonator

as LC pads separate due to finger bending, resonant frequency shifts because center to center distance changes



Springs in Series
k2 >> k1 so that effect of pip bend much greater than mcp, so can "directly" measure pip

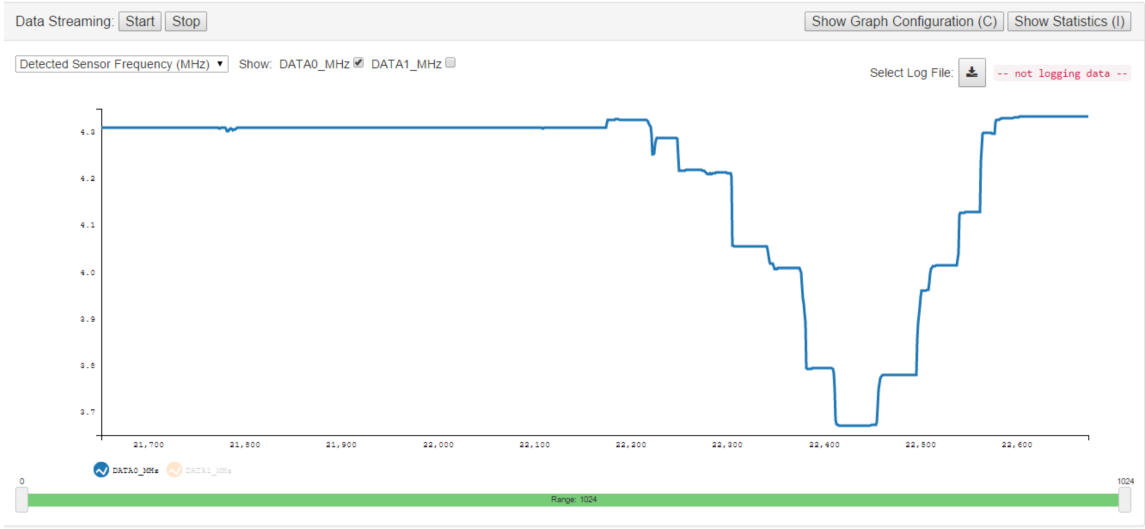


$$f_o\,(Hz) = \frac{1}{2\pi\sqrt{LC(\theta)}}$$

$$B(z) = \frac{\mu_o N_1 I_1 R_1^2}{2\left(R_1^2 + z^2\right)^{\frac{3}{2}}}$$

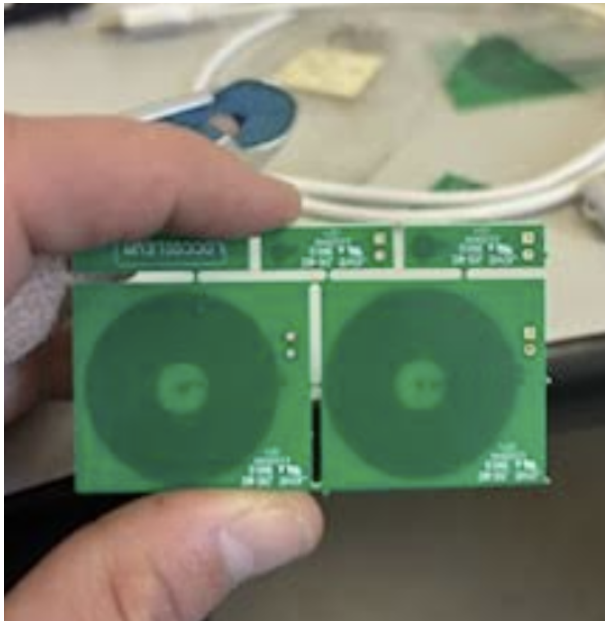
[Mutual inductance suppressed stacked inductors for passive wireless multi-parameter sensors | IEEE Conference Publication | IEEE Xplore]

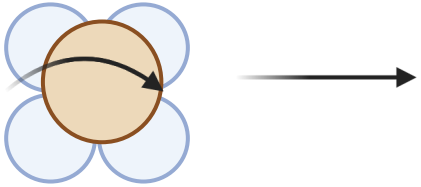
Top LC slides



Top LC Rotates

1. Overall Sensor Idea: Use mutual inductance to measure resonant frequency shifts
    a. One 2d circular coil emits a mag field and tag coil underneath gets resonated and emits resonant frequency back to top coil.
    b. If centers of coils shift, resonant frequency changes
2. arrange emitter/reader coils in 2x2 arrangement and place above 1 tag coil
    a. This will read 4 frequencies and can then detect 2d position, calibrate to flexion and abduction
3. Make use of two different stiffness material where smaller stiffness to place metal sputter coated traces to glue on arm. replace actual wires.





2D Position Sensor -> calibrate to flexion/ abduction. So 2x2 emitter/readers over 1 tag coil. 2x2 gives distribution of field over each.